

A Certified Unified Envelope for Deterministic Numerator Traces

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Abstract

We prove a single all-order geometric envelope for the Hardy-scaled deterministic numerator coefficients. If $q_* = (r_H \lambda)^{-1} = 0.7008752258547757\dots$, then $|a_n| < 48q_*^n$ for every $n \geq 2$. The constant follows from three Arb-certified residue classes of the reduced trace powers. This is a deterministic target theorem, not a uniform moving-cloud or quotient theorem.

1 Parity formula and trace residues

For odd $n \geq 3$, RH-263 gives $a_n = q_*^n / (1 + \lambda^{-n})$, so $|a_n| < q_*^n$. For even $n = 2m$,

$$a_{2m} = r_H^{-2m} \left[2 \operatorname{tr}(T^m) + \frac{x_m^2(1 - 2x_m)}{1 - x_m^2} \right], \quad x_m = \lambda^{-m}. \quad (1)$$

The endpoint fraction has modulus at most x_m^2 : indeed $x_m \leq \lambda^{-1} < \sqrt{3} - 1$ implies $|1 - 2x_m| \leq 1 - x_m^2$.

Let $m = 3k + j$, $j \in \{1, 2, 3\}$. The RH-13 nuclear certificate [1] gives

$$|\operatorname{tr} T^m| \leq \nu_j q_3^k, \quad \nu_1 < 4.623248864, \quad \nu_2 < 2.930978, \quad \nu_3 < 0.806064,$$

and $q_3 \lambda^6 < 0.801254 < 1$.

Lemma 1.1 (Residue constants). *For $m = 3k + j$,*

$$|a_{2m}| \leq (1 + 2\nu_j \lambda^{2j}) q_*^{2m}.$$

Proof. Divide (1) by $q_*^{2m} = r_H^{-2m} \lambda^{-2m}$. The endpoint term contributes at most one. The trace term is at most $2\nu_j \lambda^{2j} (q_3 \lambda^6)^k$, which is bounded by its $k = 0$ value. $\square$

2 Certified unified theorem

The 100-, 150-, and 200-decimal Arb replays certify

$$1 + 2\nu_1 \lambda^2 < 27.054, \quad 1 + 2\nu_2 \lambda^4 < 47.538, \quad 1 + 2\nu_3 \lambda^6 < 37.062.$$

Theorem 2.1 (Unified deterministic trace envelope). *For every integer $n \geq 2$,*

$$\boxed{|a_n| < 48q_*^n.}$$

Proof. The odd case has constant one. The lemma and the three strict residue bounds give the even case. $\square$

The order-2–28 atlas has observed ratios between 0.825 and 1.048, but these finite values are only a cross-check and are not used in the proof. The theorem supplies a deterministic all-order envelope, not a moving-cloud coefficient bridge or a uniform quotient tail. The obligation vector remains $(0, 0, 0, 1, 1)$ and Gates A–E remain false/open.

References

- [1] Bin Wang. Validated reduced-sector spectral gaps. *Preprint*, 2026. RH-13 preprint.